

Electroporation

Overview

Introduce foreign plasmid DNA in competent cells through exponential electrical flow in order to replicate plasmids in *B. subtilis*.

A. Materials

- a. Recovery medium
 - i. LB medium
 - ii. 0.5 M sorbitol
 - iii. 0.38 M mannitol
- b. Competent cells
 - i. *Bacillus Subtilis* 168 electrocompetent cells
 - ii. *Bacillus Subtilis* WB800 electrocompetent cells
- c. EP buffer
- d. Plasmid
- e. Electroporation cuvette
- f. 1.5mL microtube
- g. Electroporator
- h. Centrifuge
- i. LB plate with erythromycin
- j. Ice bucket with ice
- k. Water bath
- l. Cell spreader
- m. Bunsen burner

B. Procedure

Making electrocompetent cells for *B. subtilis* 168:

1. Dilute the overnight bacteria culture to 16-fold in growth medium
2. Incubate the diluted culture at 37 °C with shaking at 200 rpm until the OD₆₀₀ reaches 0.85–0.95.
3. Cool the culture on ice for 5 min.
4. Centrifuge the culture at 5,000 × g for 10 min
5. Carefully discard the supernatant and wash the cell pellet twice with ice-cold EP buffer.
6. Resuspend the pellet in 1/20 of the original culture volume of electroporation medium.
7. Aliquot the electrocompetent cells into pre-chilled microcentrifuge tubes
8. Store the electrocompetent cells at –80°C until use.

Making electrocompetent cells for *B. subtilis* WB800:

1. Dilute the overnight bacteria culture to 16-fold in growth medium
2. Incubate the diluted culture at 37°C with shaking at 200 rpm until the OD₆₀₀ reaches 2.2.
3. Cool the culture on ice for 5 min.
4. Centrifuge the culture at 5,000 × g for 10 min.
5. Carefully discard the supernatant and wash the cell pellet four times with ice-cold EP buffer with 0.5 mol/L trehalose .
6. Resuspend the cells in 1/40 of the original culture volume of ice-cold EP buffer.
7. Aliquot the electrocompetent cells into pre-chilled microcentrifuge tubes
8. Store the electrocompetent cells at -80°C until use.

Electroporation of cells:

1. Prewarm SOC Recovery medium in 37°C
2. Pre-chill 0.2 cm electroporation cuvette
3. Mix plasmid with 100 µl of electrocompetent cell mixture in at 15 ml centrifuge tube
Plasmid volume to add:
4. Set up electroporator (Gene Pulster Xcell)

<i>Bacillus subtilis</i> 168	Parameters	<i>Bacillus subtilis</i> WB800	Parameters
Voltage	2.1,2.2,2.3 kV	2.1,2.3,2.5 kV	kV
Resistance	300Ω	250Ω	Ω
Capitance	25 µF	25 µF	µF

5. Transfer the mixture of cells and DNA to a cold electroporation cuvette and flip the suspension to the bottom.
6. Place the cuvette in the shock pod, close the lid, and wait for the beep (approximately 1–2 s).
7. Immediately remove from the shock pad and add 2mL SOC medium.
8. Check and record the parameter.
9. Transfer to liquid culture tube & incubate at 37°C for 3 hrs, shaking at 200rpm
10. Centrifuge the tubes at 5000 x g for 5 minutes.
11. Plate the transformed cells onto selective LB agar plates containing erythromycin or chloramphenicol and incubate at 37°C overnight

Electroporation

operate under the BSC
unless operating
Electroporator

